N=5 Kij Structure Study: Figure Report

Scope

This report documents the four figures generated by the current analysis and plotting pipeline:

1. `analysis_overview.png`
2. `feature_profiles.png`
3. `representative_kij_gallery.png`
4. `n5_top1_verification_en.png`

All figures were produced from the result file:

- `results/data/n5_strict_results.json`

The relevant code paths are:

- `scripts/analyze_strict_results.py`
- `src/feature_analysis.py`
- `scripts/plot_strict_results.py`

Model Definition

The system is a network of `N = 5` damped, driven, nonlinearly coupled pendula:

$$\ddot{\theta}_i + \gamma\dot{\theta}_i + \omega_0^2 \sin\theta_i = \sum_{j=1}^{N} K_{ij}\sin(\theta_j - \theta_i) + \delta_{i,d} F\cos(\Omega t),$$

where:

- `d` is the driven node
- `K = (K_{ij})` is the coupling matrix
- `K_{ii} = 0`
- in this model, `K_{ij}` represents the influence from node `j` to node `i`

This directional interpretation matters for all path and strength features.

Response and Selectivity Metrics

For a fixed pair `(K, \Omega)`, the code simulates the system and extracts the steady-state Fourier response amplitude at the driving frequency:

$$A_i(K,\Omega) = \left| \frac{1}{T_{\text{tail}}} \int_{t \in \text{tail}} \theta_i(t) e^{-i\Omega t}\, dt \right|.$$

Let the set of non-driven nodes be

$$\mathcal{N}_d = \{1, \ldots, N\} \setminus \{d\}.$$

The dominant non-driven node is

$$m^*(K,\Omega) = \operatorname{argmax}_{i \in \mathcal{N}_d} A_i(K,\Omega).$$

The selectivity metric used in both the search and the final verification plot is

$$S(K,\Omega) = \frac{\max_{i \in \mathcal{N}_d} A_i(K,\Omega)}{\max_{i \in \mathcal{N}_d \setminus \{m^*\}} A_i(K,\Omega)}.$$

This is a ratio of the strongest non-driven response to the second strongest non-driven response. It is not the same as the absolute peak amplitude.

Candidate Labels Used in the Analysis Figures

Each coarse candidate is assigned one of four labels:

- `stable_high`: candidate appears in `final_verification` and remains stable under final multi-seed verification
- `unstable_peak`: candidate entered the robustness/refinement pipeline but failed before becoming a stable final winner
- `medium`: candidate did not enter the top unstable/stable pool, but its coarse best selectivity is at or above the 75th percentile of all coarse candidates
- `low`: all remaining candidates

These labels are used in `analysis_overview.png`, `feature_profiles.png`, and `representative_kij_gallery.png`.

Feature Definitions

Below, `K` denotes the `5 x 5` coupling matrix and `d` denotes the driven node. The coarse-response target node `m` is the non-driven node with the largest coarse-stage amplitude among the non-driven nodes.

Metadata and Response-Level Features

- `kidx`
  Integer identifier of the sampled matrix.

- `coarse_rank`
  Rank of the candidate after sorting all coarse candidates by `best_selectivity` in descending order.

- `best_omega`
  Coarse-stage frequency that maximizes the coarse selectivity for this candidate.

- `best_selectivity`

  Coarse-stage best selectivity:

$$\max_{\Omega \in \Omega_{\text{coarse}}} S(K, \Omega).$$

- `target_idx`

  Index of the dominant non-driven node in the coarse-stage best-amplitude vector.

- `target_amp`

  Amplitude of the dominant non-driven node at the coarse-stage best frequency:

$$A_m(K, \Omega_{\text{best}}).$$

- `second_amp`

  Amplitude of the second strongest non-driven node at the same frequency.

- `target_dominance_coarse`

  Ratio between the dominant and second dominant non-driven amplitudes:

$$\frac{\text{targetamp}}{\text{secondamp} + \varepsilon}.$$

- `coarse_target_share`

  Fraction of total non-driven amplitude captured by the target node:

$$\frac{A_m}{\sum_{i \in \mathcal{N}_d} A_i + \varepsilon}.$$

- `final_avg_selectivity`

  Mean final selectivity across all final verification seeds.

- `final_min_selectivity`

  Minimum final selectivity across all final verification seeds.

Norm and Sign Features

- `fro_norm`

  Frobenius norm of the full matrix:

$$|K|_F = \sqrt{\sum_{i,j} K_{ij}^2}\,.$$

Meaning: overall coupling magnitude.

- `asymmetry_norm`

  Frobenius norm of the antisymmetric mismatch:

$$|K - K^T|_F.$$

Meaning: absolute nonreciprocity strength.

- `asymmetry_ratio`

  Normalized nonreciprocity level:

$$\frac{|K - K^T|_F}{|K|_F + \varepsilon}\,.$$

Meaning: nonreciprocity relative to total coupling strength.

- `positive_ratio`

  Fraction of off-diagonal nonzero entries with positive sign:

$$\frac{\#\{K_{ij} > 0,\ i \neq j\}}{\#\{K_{ij} \neq 0,\ i \neq j\}}\,.$$

Meaning: sign balance of excitatory-like couplings.

- `negative_ratio`

  Fraction of off-diagonal nonzero entries with negative sign:

$$\frac{\#\{K_{ij} < 0,\ i \neq j\}}{\#\{K_{ij} \neq 0,\ i \neq j\}}\,.$$

Meaning: sign balance of suppressive-like couplings.

Magnitude Statistics

- `mean_abs_weight`
  Mean absolute off-diagonal coupling magnitude:

$$\frac{1}{M}\sum_{i \neq j}|K_{ij}|.$$

- `std_abs_weight`
  Standard deviation of the absolute off-diagonal magnitudes.

- `max_abs_weight`
  Maximum absolute matrix entry:

$$\max_{i,j}|K_{ij}|.$$

These three features describe how strong and how uneven the coupling weights are.

Node Strength Features

Because `K_{ij}` means `j \to i`, the row sum measures inflow and the column sum measures outflow.

Define

$$I_i = \sum_j |K_{ij}|, \qquad O_i = \sum_j |K_{ji}|.$$

Then:

- `drive_in_strength = I_d`
  Absolute inflow received by the driven node.

- `drive_out_strength = O_d`
  Absolute outflow emitted by the driven node.

- `target_in_strength = I_m`
  Absolute inflow received by the target node.

- `target_out_strength = O_m`
  Absolute outflow emitted by the target node.

- `target_sink_bias`
  Net sink-like tendency of the target:

$$I_m - O_m.$$

Positive values mean the target behaves more like an absorber/sink than a broadcaster/source.

Direct and Multi-Hop Path Features

- `target_direct_from_drive`
  Direct drive-to-target link strength:

$$|K_{md}|.$$

Meaning: one-step directional coupling from the driven node to the target.

- `target_path_2hop`
  Absolute two-hop drive-to-target path score:

$$P_m^{(2)} = \sum_{j \neq d,\, m} |K_{jd} K_{mj}|.$$

Meaning: total strength of all two-step directed paths from the drive to the target.

- `target_path_advantage_2hop`
  Target two-hop score normalized by the strongest competing non-driven node:

$$\frac{P_m^{(2)}}{\max\limits_{r \in \mathcal{N}_d,\ r \neq m} P_r^{(2)} + \varepsilon}.$$

Meaning: whether the target has a stronger two-hop route than other candidates.

- `target_path_3hop`
  Absolute three-hop drive-to-target path score:

$$P_m^{(3)} = \sum_{j \neq d,\, mk \neq d,\, m,\, j} |K_{jd} K_{kj} K_{mk}|.$$

Meaning: total strength of all three-step directed paths from the drive to the target.

- `target_path_advantage_3hop`
  Target three-hop score normalized by the strongest competing node:

$$\frac{P_m^{(3)}}{\max\limits_{r \in \mathcal{N}_d,\ r \neq m} P_r^{(3)} + \varepsilon}.$$

Figure 1: `analysis_overview.png`

Purpose

This figure provides a compact overview of the search outcome and the first layer of feature screening.

Plot Construction

It is a `2 x 2` panel figure:

Top-left: coarse selectivity histogram

- Input data: all coarse candidates
- Quantity plotted: `best_selectivity`
- Plot type: histogram
- Additional markers:
- vertical line at the 75th percentile

- vertical line at the 90th percentile

Meaning: shows how rare high coarse selectivity is and how extreme the upper tail is.

Top-right: label counts

- Input data: all coarse candidates after label assignment
- Plot type: bar chart
- Categories:
- `stable_high`
- `unstable_peak`
- `medium`
- `low`

Meaning: gives the class balance of the entire run.

Bottom-left: asymmetry ratio vs coarse selectivity

- x-axis:

$$\text{asymmetryratio} = \frac{|K - K^T|_F}{|K|_F + \varepsilon}$$

- y-axis: `best_selectivity`
- Color: candidate class label

Meaning: shows whether stronger nonreciprocity alone explains coarse selectivity.

Bottom-right: most discriminative structural features

- Input data: per-label feature means over all labeled candidates
- Score used for each feature `f`:

$$\text{score}(f) = \frac{\max_{\ell} \mu_{\ell,f} - \min_{\ell} \mu_{\ell,f}}{\sigma_f + \varepsilon}$$

where: - `\mu_{\ell,f}` is the mean of feature `f` inside label `\ell`
- `\sigma_f` is the global standard deviation of feature `f`

- Plot type: horizontal bar chart of the top-scoring features

Meaning: this panel identifies which structural quantities change the most across the sample classes, relative to their overall variability.

Figure 2: `feature_profiles.png`

Purpose

This figure gives a more direct visual comparison of the top-ranked structural features across the sample classes.

Plot Construction

- The code first sorts all structural features by the discriminative score defined above.
- It then selects the top four features.
- The figure is a `2 x 2` panel layout, one panel per feature.

Each panel uses:

- x-axis: sample class (`stable_high`, `unstable_peak`, `medium`, `low`)
- y-axis: feature value
- points: jittered scatter of all samples in the class
- black horizontal segment: mean feature value in that class

Meaning: this figure is intended to reveal whether the top-ranked structural features show class separation, overlap, outliers, or broad spread.

Figure 3: `representative_kij_gallery.png`

Purpose

This figure shows representative coupling matrices for the main sample classes.

Plot Construction

The gallery contains one matrix per class:

- `Best stable`
- `Best unstable`
- `Best medium`
- `Best low`

Selection rule:

- `Best stable`: highest `final_avg_selectivity`
- `Best unstable`: highest `best_selectivity`
- `Best medium`: highest `best_selectivity`
- `Best low`: highest `best_selectivity`

Each panel is:

- a heatmap of the full `5 x 5` matrix `K`
- plotted with a diverging colormap centered at zero
- scaled symmetrically with

$$v_{\max} = \max_{i,j} |K_{ij}|$$

Meaning:

- red: positive coupling
- blue: negative coupling
- pale/white: weak coupling

This figure is meant for qualitative visual comparison of structural motifs across classes.

Figure 4: `n5_top1_verification_en.png`

Purpose

This is the detailed verification figure for the strongest verified winner of the full run.

Candidate Selection

The current code always chooses the strongest verified candidate:

$$\mathrm{argmax}_{c \in \mathrm{finalverification}} \mathrm{avgselectivity}(c).$$

If no final candidates exist, the script falls back to `refined_results`, then `robustness_test`, then `coarse_results`.

Plot Construction

The figure has five panels arranged over three rows.

Row 1, left: amplitude-frequency plot

- Input: selected winner `K`
- Frequency grid: `omega_fine`
- Seed: first final seed
- Plotted quantity:

$$A_i(K, \Omega)$$

for all nodes `i`

Additional elements:

- dashed vertical line at the frequency where the recomputed selectivity peaks

- annotation showing the peak selectivity value

Important note: the annotated "Best" refers to peak selectivity, not maximum absolute amplitude.

Row 1, right: zoom near best frequency

- Same quantity as above
- Restricts the x-range to a window centered on the best frequency

Meaning: local view of the resonance structure near the selected operating point.

Row 2, left: seed robustness bar chart

- x-axis: node index
- y-axis: Fourier amplitude at the selected best frequency
- bars: one bar group per final verification seed

Meaning: tests whether the target-node dominance persists across seeds.

Row 2, right: selectivity vs Omega

- x-axis: `\Omega`
- y-axis:

$$S(K, \Omega)$$

- input: recomputed fine frequency sweep for the chosen winner

Additional elements:

- dashed vertical line at the best operating frequency
- optional shaded target band if target-band mode is enabled

Meaning: this panel shows directly how sharply the chosen matrix concentrates response into a single non-driven node as the drive frequency changes.

Row 3: time series

- x-axis: time
- y-axis: angle `\theta_i(t)`
- plotted for all nodes at the selected best frequency and one seed

Meaning: gives a time-domain view of the winner's dynamical behavior.

Practical Reading Guide

If the goal is to understand what makes a `K` successful, the recommended reading order is:

1. `analysis_overview.png` - especially the bottom-right feature-importance panel
2. `feature_profiles.png` - to inspect overlap and separation by class
3. `representative_kij_gallery.png` - to visually compare matrix motifs
4. `n5_top1_verification_en.png` - to inspect the final winning matrix in detail

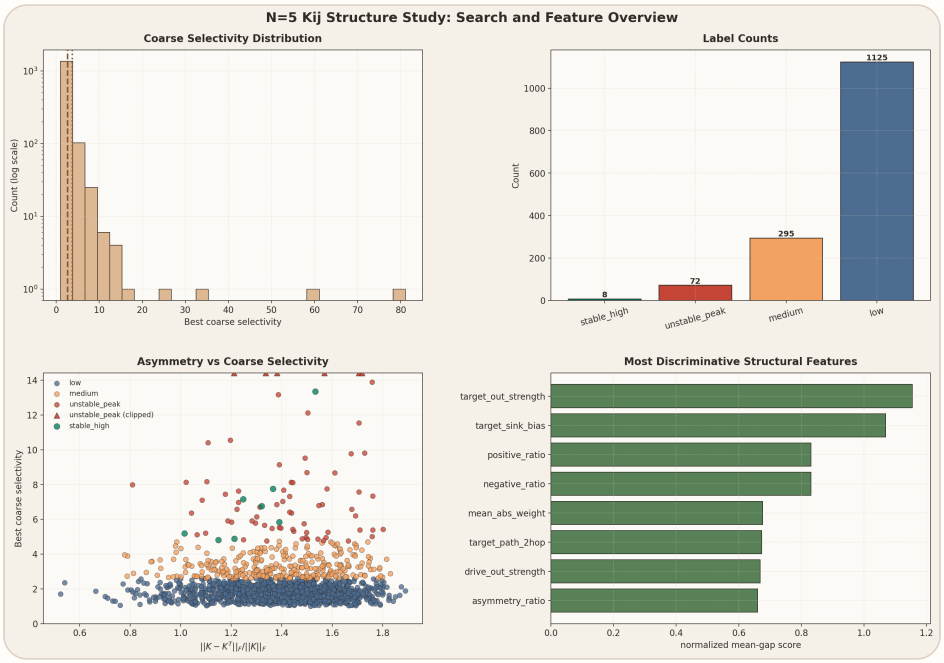
File References

- `results/figures/analysis_overview.png`
- `results/figures/feature_profiles.png`
- `results/figures/representative_kij_gallery.png`
- `results/figures/n5_top1_verification_en.png`

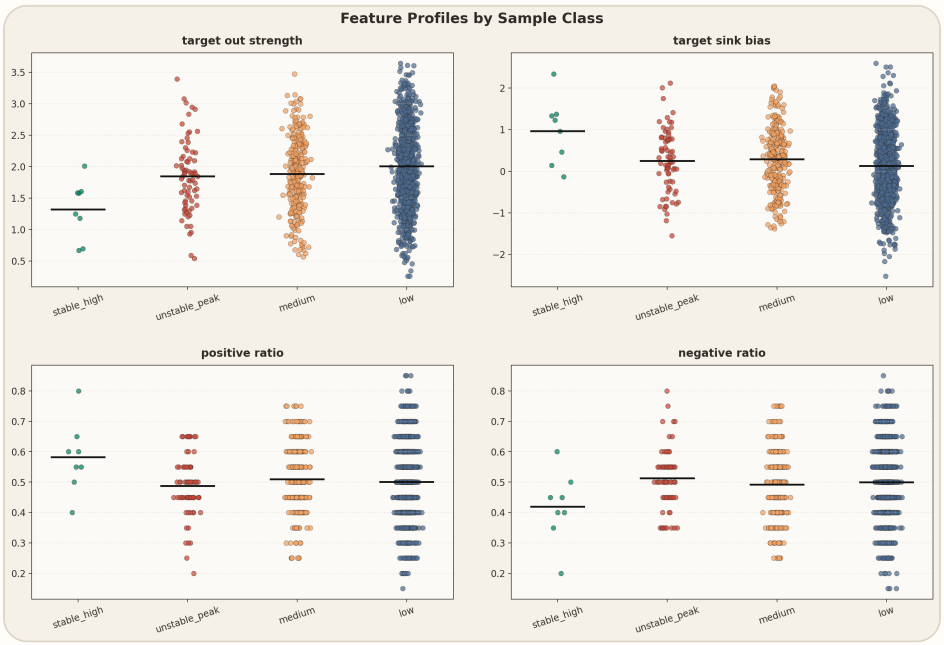
Rendered Figures

The images below are embedded directly from the local results directory.

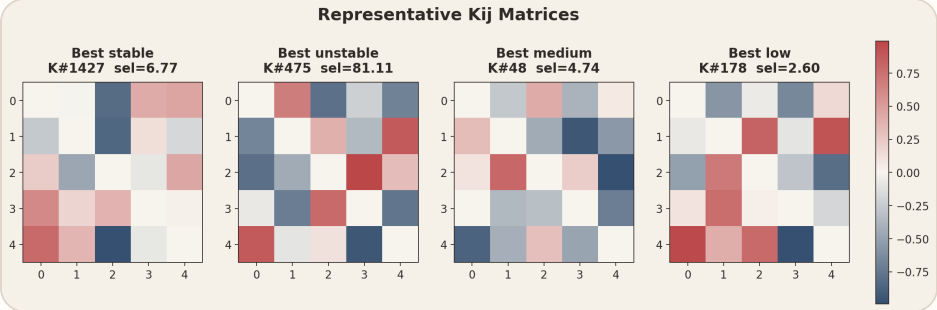
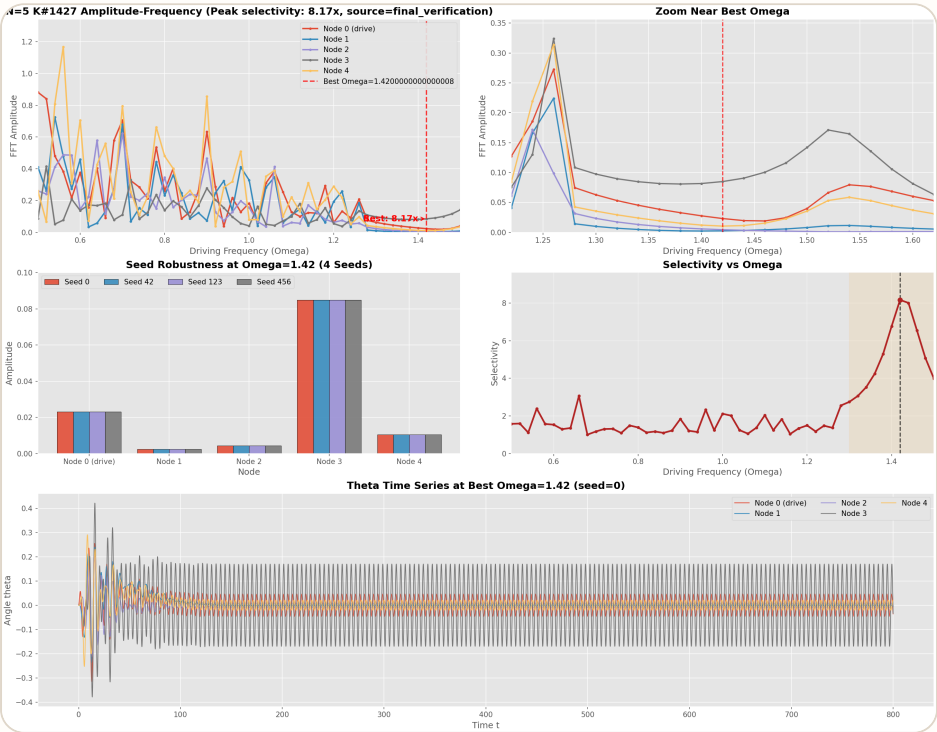
Analysis Overview



Feature Profiles



Representative Kij Gallery



Top-1 Verification